

Exp: 07

Write the Python Program to implement BFS

Aim:

To write a python program to implement BFS.

Objective:

To understand level-order traversal of a graph using a queue-based (FIFO) approach

Algorithm:

Step 1: Initialize a queue data structure and a set to keep track of visited nodes.

Step 2: Add the starting node to the queue and to the set of visited nodes.

Step 3: While the queue is not empty, dequeue the next node from the queue.

Step 4: For each neighbor of the dequeued node, if it has not been visited yet, add it to the queue and mark it as visited.

Step 5: Repeat steps 3-4 until the queue is empty

Program:

```
from collections import deque
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
def bfs(graph, start):  
    visited = set()  
    queue = deque([start])  
  
    visited.add(start)
```

```
while queue:
    node = queue.popleft()
    print(node, end=" ")

    for neighbor in graph[node]:
        if neighbor not in visited:
            visited.add(neighbor)
            queue.append(neighbor)
```

```
bfs(graph, 'A')
```

Output:

A B C D E F

Result:

Thus, the was executed and found the output was found to be correct.